

PRASUNETHON 2.0

Project Title: *OpteraAI*

Domain: Artificial Intelligence & Machine Learning

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Problem Statement:

Students and professionals work with information using several independent applications like email, calendar, documents, meetings, job portal and educational portal.

Key Problems

- Important emails and opportunities get missed.
- The user must categorize and prioritize the information manually.
- The deadlines and meetings have to be tracked separately.
- Information such as documents, resumes, and applications has to be handled separately.
- Repeated filling of forms wastes time.
- Current AI systems only assist users but cannot carry out full processes.
- During meetings, the user cannot constantly track and answer queries.

Core Problem

There is no single system that can comprehend information, make decisions and automatically carry out productivity tasks in several applications.

Proposed Solution:

OpteraAI is an integrated AI productivity system created using 16 AI agents specifically designed to work in collaboration on one dashboard.

How it works:

- Interpret incoming information from Gmail, documents, and meetings.
- Categorize and prioritize relevant information.
- Perform research to get deadlines, opportunities, and needed actions.
- Organize information related to knowledge, documents, meetings, and schedule.
- Automatically act based on applying, scheduling, alerting and updating information.
- Learn from users to offer personalized career and educational support.

Unique Aspect

Unlike other systems that focus on isolated AI features, OpteraAI combines various agents to perform end-to-end workflows autonomously.

Features

Intelligent Email Processing

- Emails are automatically analyzed and classified based on priorities. Important opportunities, deadlines and action items are recognized without the need for manual sorting of emails by the user.

Automated Applications

- The Application Agent can complete supported application forms using information available from the user's profile and resume, thus saving data entry efforts.

Document and Knowledge Management

- Documents and email attachments are analyzed and managed as personal knowledge, which is searchable. Users can ask questions and get information from their own documents.

Autonomous Meeting Agent

- The Meeting Agent can attend the meetings authorized by the user on his/her behalf. Using the context of the user, it can answer questions in the boundaries of its instructions.

Career and Learning Assistance

- Career and Learning Agents provide suggestions, highlight skill gaps and help create learning and career plans for individuals.

Smart Scheduling

- Calendar and Notification Agents schedule meetings, set deadlines and reminders.

Idea/Approach

Multi-agent Architecture:

OpteraAI uses various AI agents for complex productivity work, unlike the use of only a single AI agent.

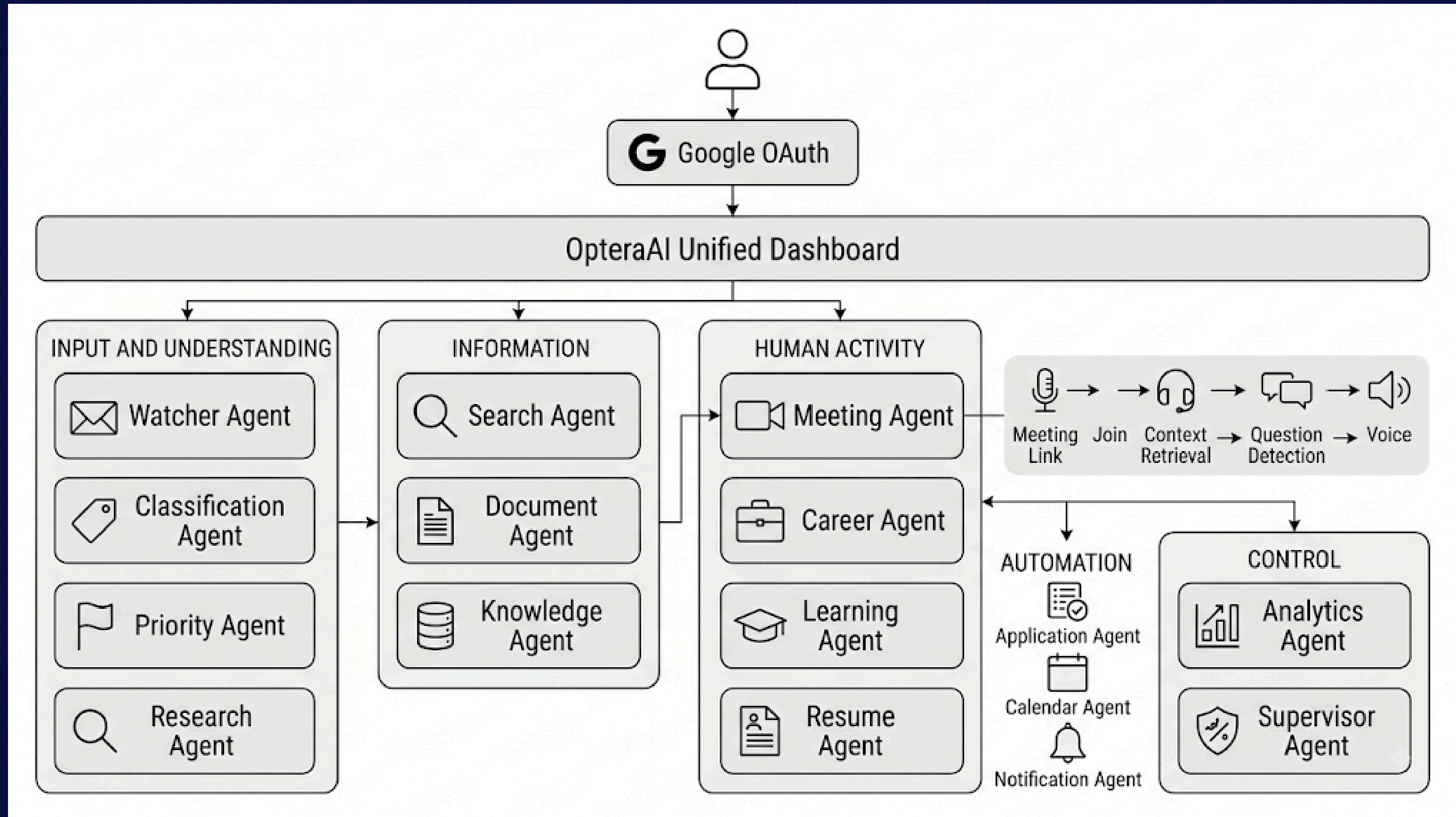
Tech Stack:

- **Frontend:** React.js, Vite, TypeScript, Tailwind CSS
- **Backend:** Python, FastAPI, SQLAlchemy
- **AI:** Groq API, Gemini API, Sentence Transformers
- **RAG:** ChromaDB, Vector Embeddings, Cosine Similarity
- **Automation:** Playwright
- **Meetings:** Whisper, FFmpeg
- **Documents:** PyMuPDF, pdfplumber, python-docx, OCR
- **Database:** PostgreSQL
- **Authentication:** Google OAuth, JWT, Gmail API
- **Scheduling:** AP Scheduler

Approach:

Input → Understand → Classify → Prioritize → Research → Execute → Schedule → Notify → Analyze

System Architecture / Workflow



Innovation & Existing Solutions

Productivity apps that exist nowadays concentrate their efforts on a single aspect, like email, calendar, meetings, documents, or career management. Users are still required to transfer data from one application to another and do a lot of tasks manually.

The concept of **OpteraAI** is to unify these processes by means of specially developed agents that can exchange data between them.

What makes it unique?

- The data exchange between the agents can be performed automatically.
- The personal documents and context are taken into account when making decisions.
- Tasks can be automated, not just suggested.
- Calendar and notifications workflows are unified.
- The Meeting Agent can act on behalf of the user according to a given context and instructions.
- All agents are accessible from a single dashboard.

The main concept is not an AI personal assistant, but the whole AI system capable of processing the data and performing the further workflow.

Impact & Future Scope

Impact

OpteraAI can reduce the time spent on repetitive digital tasks such as checking emails, organizing information, filling applications, managing deadlines and searching through documents.

It can also help users avoid missing opportunities by connecting email monitoring, priority detection, calendar events and notifications.

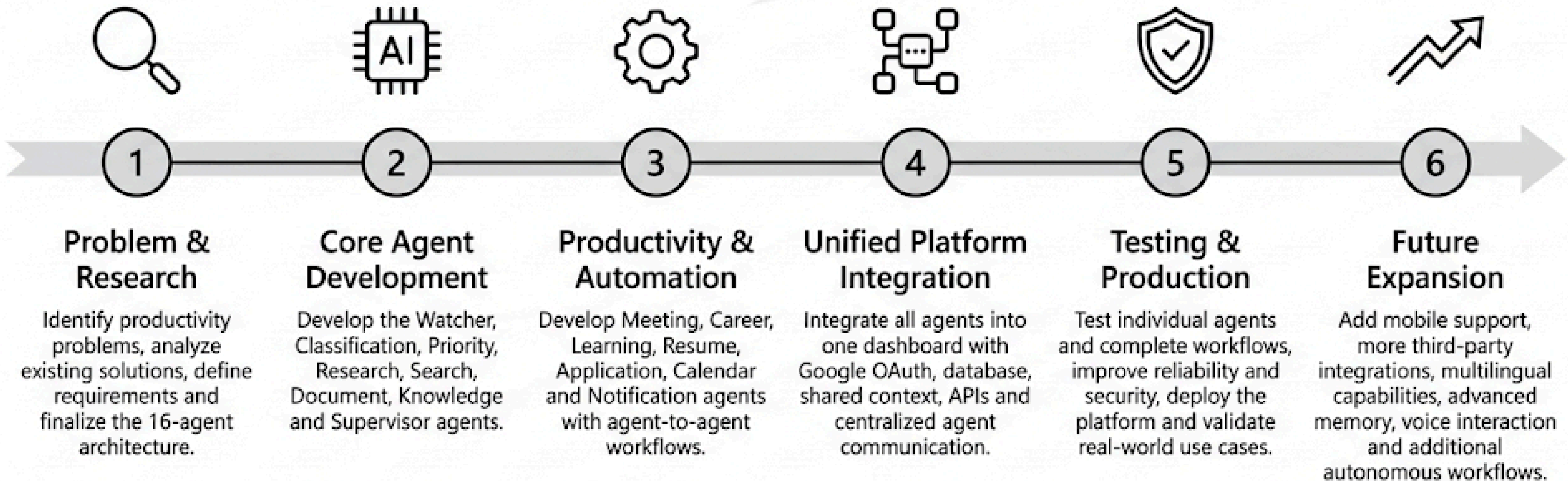
For students, the platform can bring internships, hackathons, learning, career preparation and academic activities into one place.

Future Scope

The system can be extended with mobile applications, more email and meeting platforms, additional application portals, voice interaction, multilingual support and more advanced autonomous workflows.

The Meeting Agent can also be extended to handle different meeting roles and respond using a larger personal knowledge base while following user-defined rules.

Project RoadMap



Idea → Build → Integrate → Deploy → Scale

Thank You!

OPTERA AI

Autonomous Multi-Agent AI Productivity Platform

GitHub Repo URL: [LINK](#)

Demo Video: [LINK](#)